

The Fundamentals of Programming

For Loops



Objectives

What you will learn:

1. How to repeat actions using loops.
2. The different ways that loops can be used to solve complex problems.
3. To diagnose and fix problems with loops.

The For Loop

Where while loops are usually used when you're not exactly sure when the loop will end, for loops are for when you know how many times you want to loop.

Basic for loop:

```
for count in range(start, end):  
    #do stuff here
```

Example 1:

```
for count in range(0, 10):  
    print(count)
```

Example 2:

```
for i in range(0, 10):  
    print('Hi')
```

A Deeper Look

The variable

A **for loop** is a bit more complicated than a while loop.

First, it uses a function called **range** which takes a **start** and **end** value.

Second, it has a variable which tracks how many times you've looped.

```
for var in range(0, 10)
```

var – used to tell how many times you've looped

range – a function that gives the start and end value of the loop.

0, 10 – the start and end values.

For loops

A walkthrough

There are actually three ways to write a **for loop**.

```
for a in range(0, 3):  
    print(a)
```

Output

0
1
2

```
for a in range(3):  
    print(a)
```

Output

0
1
2

```
for a in range(3, 0, -1):  
    print(a)
```

Output

3
2
1

For loops

Practice

With each of the following programs, state the output.

```
a = 0
for g in range(5):
    a = a + g
print(a)
```

```
for var in range(0, 4):
    print('Hi')
print(var)
```

```
x = 2
for g in range(x, 6):
    print(x * 2)
```

For loops

Practice

What method would you use to determine the outcome?

```
x = 0
y = 0
for count in range(5):
    x = count + y
    y += 1
    print(x, y, count)
```

count	x	y	output
	0	0	
0	0	1	
1	2	2	
2	4	3	
3	6	4	
4	8	5	4 8 5

Recap

We learned

1. How to create a for loop using the range function.
2. The ways that a for loop differs from a while loop, and how it's used when we know how many times we want to loop.
3. There are three different ways to create for loop: using a single integer, using a start and end, and with a start, end and increment.